

Step	Step Description	Agents Involved	Tool Calls and Parameters	Reason
1	User asks: Which buildings in Exeter have green roofs?	human, host_agent	Message sent from human to host_agent	User initiates the query.
2	Plan steps to answer the query	planning_agent	send_message to planning_agent with task_description	planning_agent is chosen to break down the geospatial query into actionable steps, as per its system prompt.
3	Planning agent outputs steps: ["Find Exeter 1 area", "Search for buildings in Exeter with green roofs as many search results"]	planning_agent	Output steps	Steps are based on the prompt and reasoning template for geospatial queries.
4	Find the area polygon for Exeter	host_agent, named_area_agent	send_message to named_area with task_description: 'Find the area polygon for Exeter.'	named_area_agent is chosen because the prompt specifies that named_area contains polygons of areas such as cities.
5	Retrieve named area polygons (including Exeter)	named_area_agent	call_os_ngd with polygon_or_point=True, filename='exeter_area_polygon'	The named_area_agent uses the OS NGD tool to get all named area polygons, as per its API capabilities.
6	Filter to the specific polygon for Exeter	host_agent, data_analysis_agent	send_message to data_analysis_agent with task_description: Filter 'exeter_area_polygon' to find the specific polygon for Exeter. Save as 'filtered_exeter_area_polygon'.	Filtering is needed because the initial artifact contains multiple polygons; only the one for Exeter is relevant.
7	Generate metadata for 'exeter_area_polygon'	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['exeter_area_polygon']	Metadata is generated to understand the columns and structure for accurate filtering.
8	Filter code execution: select polygons where name1_text contains 'Exeter' and descriptiongroup is one of ['Settlement', 'Part Of Settlement', 'City', 'Town', 'Suburban Area']	coding_agent (via code_executor)	Code uses columns 'name1_text' and 'descriptiongroup' as per metadata; output is 'filtered_exeter_area_polygon'	These columns are chosen based on metadata and the need to accurately identify the Exeter city polygon.
9	Confirm filtered polygon for Exeter	data_analysis_agent	Output: 1 polygon for Exeter in 'filtered_exeter_area_polygon'	Confirms that the filtering step was successful.
10	Search for buildings with green roofs within Exeter area	host_agent, buildings_agent	send_message to buildings with task_description: Search for buildings with green roofs within 'filtered_exeter_area_polygon'	buildings_agent is chosen because the query is about buildings, and the bbox parameter is set to the filtered Exeter polygon artifact.
11	Retrieve all buildings within Exeter area	buildings_agent	call_os_ngd with bbox='filtered_exeter_area_polygon', filename='all_buildings_exeter_green_roof_search'	The buildings_agent uses the OS NGD tool to get all buildings within the Exeter area; filters are not applied at this stage as green roof is a property to be filtered later.
12	Filter buildings for green roofs	host_agent, data_analysis_agent	send_message to data_analysis_agent with task_description: Filter 'buildings_all_buildings_exeter_green_roof_search' for green roofs, save as 'buildings_exeter_green_roofs_only'	Filtering is required because the initial artifact contains all buildings; only those with green roofs are relevant.
13	Generate metadata for 'buildings_all_buildings_exeter_green_roof_search'	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['buildings_all_buildings_exeter_green_roof_search']	Metadata is generated to identify relevant columns for green roof filtering.
14	Filter code execution: select buildings where 'roofmaterial_greenroofpresence' == 'Present' or 'roofmaterial_primarymaterial' == 'Green Roof'	coding_agent (via code_executor)	Code uses columns 'roofmaterial_greenroofpresence' and 'roofmaterial_primarymaterial' as per metadata; output is 'buildings_exeter_green_roofs_only'	These columns are chosen based on metadata and the need to accurately identify buildings with green roofs.
15	Confirm filtered buildings with green roofs	data_analysis_agent	Output: 225 buildings found in 'buildings_exeter_green_roofs_only'	Confirms that the filtering step was successful.
16	Plot buildings with green roofs and overlay Exeter area boundary	host_agent, plotting_agent	send_message to plotting_agent with task_description: Plot 'buildings_exeter_green_roofs_only' and overlay 'filtered_exeter_area_polygon'	plotting_agent is chosen to visualize the results; both artifacts are used for plotting as per the prompt.
17	Generate metadata for plotting	plotting_agent	generate_metadata_for_artifacts with artifact_names=['buildings_exeter_green_roofs_only', 'filtered_exeter_area_polygon']	Metadata is generated to ensure correct columns and CRS for plotting.
18	Plotting code execution: create folium map, convert CRS to EPSG:4326, drop datetime columns, plot Exeter boundary and green roof buildings, save as HTML	coding_agent (via code_executor)	Code ensures compatibility with folium by dropping datetime columns and converting CRS; outputs map file and summary	These steps are necessary for folium plotting and are based on the plotting agent's template and metadata.
19	Confirm map generation and provide summary	plotting_agent	Output: Map file 'exeter_green_roof_buildings_map.html' and summary of first 5 buildings	Confirms that the plotting step was successful and provides a preview.
20	Final output to user	host_agent	Output: 225 buildings with green roofs in Exeter, map generated, summary provided	Final step to communicate results to the user, as per the host_agent's role.

Overall Summary: - Databases used: Named area (for Exeter polygon), Buildings (for all buildings in Exeter), Data analysis (for filtering), Plotting (for visualization). - Filters/code: Filtering for Exeter polygon used 'name1_text' and 'descriptiongroup' columns; filtering for green roofs used 'roofmaterial_greenroofpresence' and 'roofmaterial_primarymaterial' columns. Plotting required CRS conversion and removal of datetime columns for folium compatibility. - Final outcome: 225 buildings in Exeter were identified as having green roofs, and a folium map was generated showing their locations overlaid with the Exeter area boundary. - Overall reasoning: Each agent and tool was chosen based on its system prompt and the requirements of the step. Metadata was used to select the correct columns for filtering and plotting, ensuring accurate and relevant results at each stage.